CMT 316
HUMAN COMPUTER INTERFACE

Course Overview

- This course is an introduction to the fundamentals of human-computer interaction, user interface design, and usability analysis.
- Students will learn principles and guidelines for usability, quantitative and qualitative analysis methods, and apply them through critiques of existing interfaces and development of new ones.
- Topics covered will also include cognitive models, task analysis, psychology, experimental design, and prototyping and evaluating methods.

Learning Outcomes

- Critically discuss common methods in the user-centered design process and the appropriateness of individual methods for a given problem.
- Use, adapt and extend classic design standards, guidelines, and patterns.
- Employ selected design methods and evaluation methods at a basic level of competence.
- Build prototypes at varying levels of fidelity, from paper prototypes to functional, interactive prototypes.

Intended Skill Outcomes

- To be able to apply the principles, tools and techniques of HCI to the development of usable digital information systems, including:
  - Requirements capture
  - System design / application of design guidelines
  - Interface evaluation

Aim-Approach for this course

- Learn the major principles of HCI/Interaction
  - Usability
  - Affordance
  - Cognetics
- Learn how people think, react, acquire
  - Perception
  - Cognetics
- Learn how to design and evaluate a system
  - Development track
  - Prototyping, Evaluating
  - Research based approach

Teaching Methods:

- The course will be based on the following teaching and learning activities:
  - Lectures covering the theoretical part using Power Point presentations, Black Board, Videos/Audio … etc.
  - Review questions at the beginning of the lecture and at the end of the lecture.
  - In-Class Group Activities
  - Class Discussions with Case studies

Assessment Tools

- Exams
- Presentation
- Projects
- Discussion
- In-Class Activity
- Feedback at the end of class

Assessments

- Quiz 1 – 10%
- Quiz 2 – 10%
- In-Class Group Activity – 20 (Max 3 students)
- End-Exam – 60%

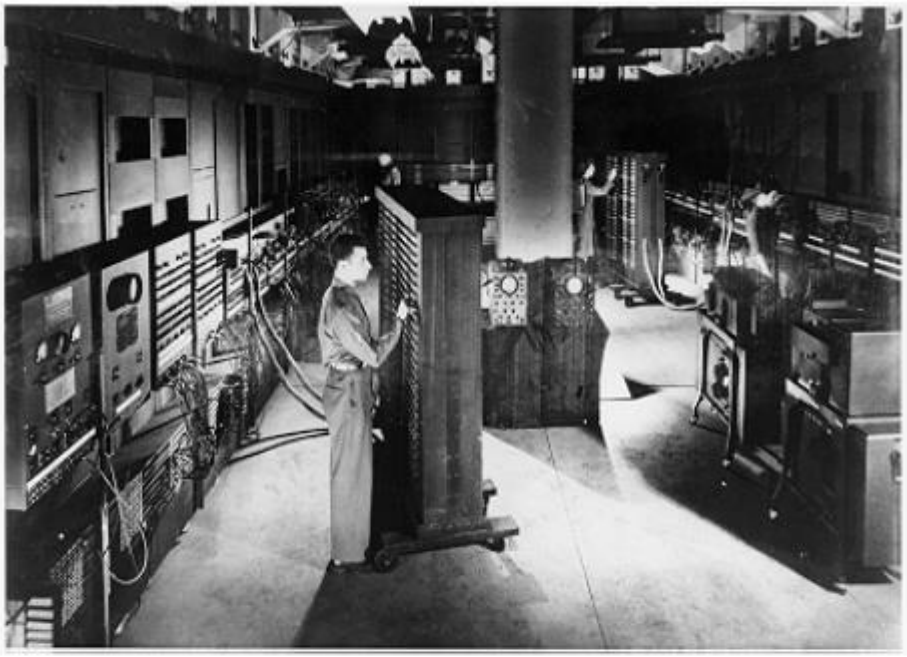
Books and Web Resources

- My Lecture Notes (ppt)
- Web Resource (Nielsen Norman Group)
  - https://www.nngroup.com/
- Book
  - Human Computer Interaction by Alan Dix, Janet Finlay et.al

Outline

- Early Human Computer Interaction
- Development of HCI
- Interaction Paradigms
- Types of user interfaces
- Disciplines contributing to HCI
- Aim-Approach for this course

Computing in 1943



ENIAC (1943) - the first electronic numerical integrator and computer in the US

Computing in 1945

- Harvard Mark I
  - ASCC: IBM Automatic Sequence Controlled Calculator (aka Mark 1)
- 55 feet long, 8 feet high, 5 tons

First game on PDP (Programmed Data Processor) in 1960

Early Human Computing Interaction

Early Human Computing Interaction

What Interactions Did We See?

What Interactions Did We See?

- Mechanical
- Poor feedback
- Specialist use
- Process control
- Calculations
- No intention to address the mass market

Computing in 1980

Computing in 2000

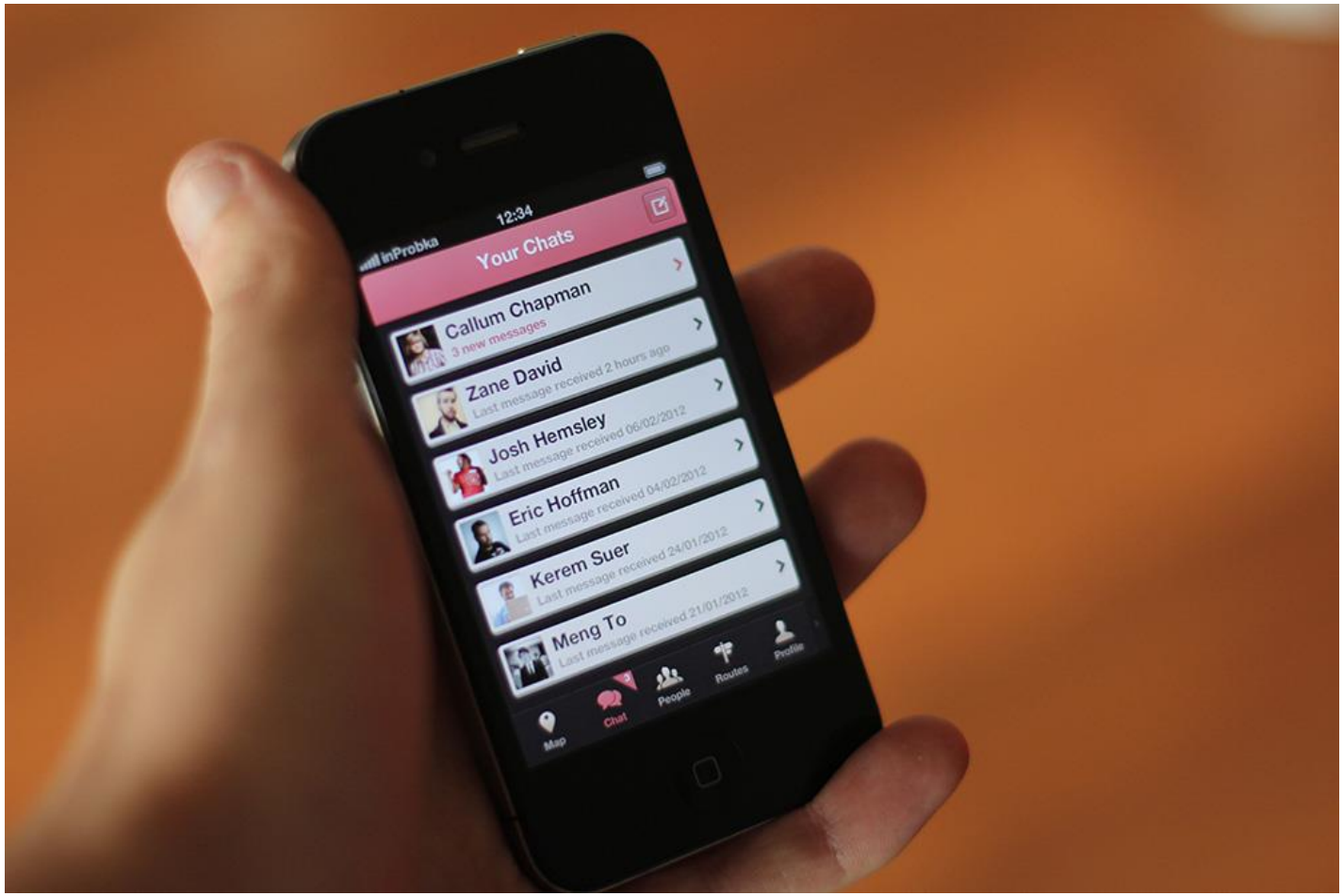
Gaming Interface in 1990

Late Human Computing Interaction

Today Human Computing Interaction

Today Human Computing Interaction

Today Human Computing Interaction



What are the Development in HCI?

Development of HCI

- Early computers (1940): extremely difficult to use
  - Large and Expensive
  - By comparison: “people time” (labour) cheap
  - Used by specialists
  - No knowledge about how to make use easier
- Late computers (1990): easy to use
  - None of these conditions hold
  - Development of PC’s major landmark
- Today (2015)-Shift to other interaction paradigms
  - **Small-handheld-Devices**
  - **Mobility**

Interaction Paradigms

- Batch Processing
- Timesharing
- Networking (1972 1st email)
- Graphical display
- Microprocessor
- WWW
- Grid/Clouds Computing
- Human Robot Interaction
- Tablet/Table Top Computing

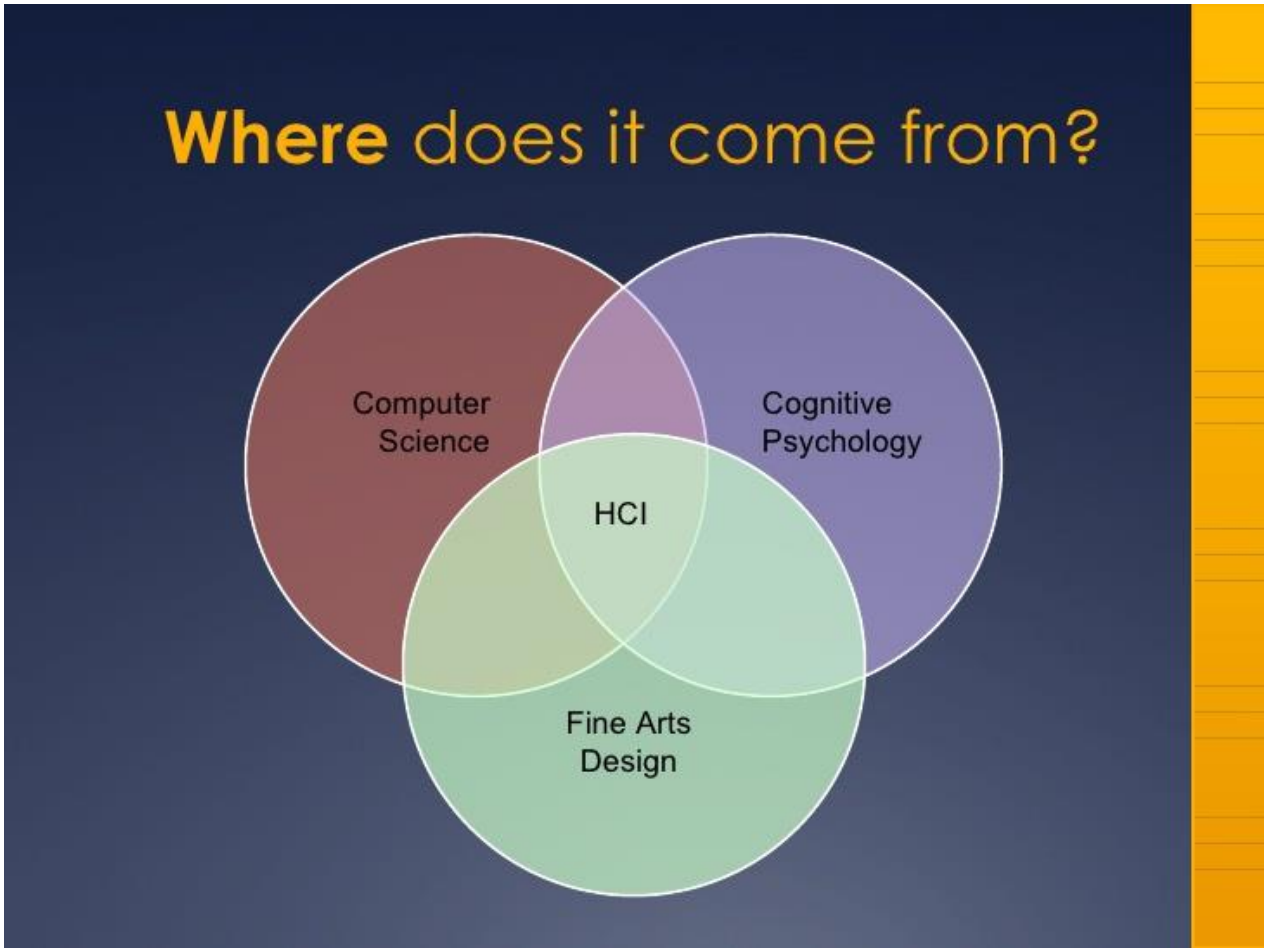
- 1950's
- 1960's
- 1970's
- 1980's
- 1990's
- 1995's
- This era …

What are the types of user interface?

Types of user interfaces

- Command Line Interface
- Menu Driven Interface
- Graphical User Interface
- Natural Language Interface

Disciplines contributing to HCI



Disciplines contributing to HCI

Aim-Approach for this course

- Learn the major principles of HCI/Interaction
  - Usability
  - Affordance
  - Cognetics
- Learn how people think, react, acquire
  - Perception
  - Cognetics
- Learn how to design and evaluate a system
  - Development track
  - Prototyping, Evaluating
  - Research based approach